

TINON TURJA MAJUMDER

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RESEARCH INTERESTS

- Data-driven and hybrid (physics + machine learning) modelling of industrial processes, with uncertainty quantification, toward digital twins of batch manufacturing.
- Learning-based decision-making and closed-loop control of industrial processes under uncertainty, using large-scale plant and sensor data.
- Physics-informed and scientific machine learning for parameter identification in reaction-sorption systems.

EDUCATION

Master of Science in Computer Science and Engineering

Jan 2024 – Aug 2025

United International University, Dhaka, Bangladesh. CGPA 3.61/4.00. Concentration: Data Science.

Relevant coursework: Machine Learning, Deep Learning, Advanced Artificial Intelligence, Computational Intelligence, Data Mining, Big Data, Image Processing, Natural Language Processing, Internet of Things, Research Methodology.

Bachelor of Science in Textile Engineering (Dyes and Chemicals)

Jan 2018 – Dec 2023

Bangladesh University of Textiles, Dhaka, Bangladesh. CGPA 3.05/4.00. Undergraduate thesis graded 4.00/4.00.

Quantitative coursework: Differential and Integral Calculus, Linear Algebra, Ordinary Differential Equations, Vector Analysis, Complex Variables, Laplace Transforms, Probability and Statistics, Computer Programming.

RESEARCH EXPERIENCE

Lead Research Assistant — SMART DYEING

Mar 2026 – Present

Asian Development Bank funding through BIRDI (Bangladesh Industry Research, Development and Innovation) under SICIP; BDT 2.18 crore (≈US\$177,000). PI: Prof. Dr. Mohammad Forhad Hossain. AI-driven closed-loop process control to reduce chemical, water and energy use in industrial knit-fabric dyeing.

- Built the project data pipeline: a verified recipe corpus of 47,403 production dyeing batches (2021–2026; 137,096 chemical line items) derived from 459,267 raw ERP records, plus 98,012 dyeing-controller batch records and 97,655 decoded machine telemetry logs.
- Carried out on-site instrumentation at the DBL Group Fabric Division dyeing facility, including installation of a sensor network on production dyeing machines.
- Designed the methodology and developed predictive models for shade-category and recipe-level resource optimisation.
- Set up laboratory trial apparatus for controlled validation experiments.
- Recognised in a Letter of Appreciation from DBL Group Fabric Division, which described the delivered baseline characterisation as “the most comprehensive, verified and operationally actionable on record for their dyeing division.”

Proposal Development and Competitive Review — SMART DYEING

Sep 2025 – Feb 2026

- Authored and defended the project proposal through four consecutive competitive BIRDI review rounds to funding approval (BDT 2.18 crore, ≈US\$177,000).

Research Assistant — Dyes and Chemicals Engineering Laboratory

Jan 2024 – Dec 2025

Bangladesh University of Textiles.

- Designed an inverse physics-informed neural network (PyTorch) constrained by pseudo-second-order kinetics to recover rate constants from batch adsorption data, and a mass-balance-coupled isotherm-kinetic surface model.
- Developed the machine-learning and kinetic modelling for two biofilm dye-adsorption manuscripts, including Gaussian process regression, leave-one-out model benchmarking and a preregistered simulation of kinetic-law identifiability.

Intern — Fakir Knitwears Ltd.

Mar 2023 – May 2023

- Observed industrial dyeing operations, quality control and large-scale production; this experience motivated the SMART DYEING project.

PREPRINTS AND MANUSCRIPTS UNDER REVIEW

[1] Md Shajjad Khan Faisal, **Tinon Turja Majumder***, Kazi Sirajul Islam*, Sk. Mainuddin, Mohammad Forhad Hossain, Niger Sultana, Mahmudul Hasan. *Interfacial Mechanisms of Reactive Dye Adsorption on a Waste-Cotton Derived PVA/TiO₂/Cellulose Biofilm: A Molecular Dynamics and Physics-Informed Predictive Framework*. ChemRxiv, 2026. doi.org/10.26434/chemrxiv.15008049/v1. Under review at *Chemical Engineering Journal Advances* (Elsevier). Role: inverse PINN modelling and code.

[2] Sk. Mainuddin, **Tinon Turja Majumder***, Md Shajjad Khan Faisal. *From Prediction to Process Control: A Critical Review of Machine Learning in Reactive Dyeing of Cotton*. ChemRxiv, 2026. doi.org/10.26434/chemrxiv.15009673/v2. Under review at *Chemical Engineering Journal Advances* (Elsevier). Role: literature screening, coding and synthesis.

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SOFTWARE AND DATASETS

- **WCF_Biofilm_Adsorption** — code and analysis for preprint [1]; Zenodo DOI 10.5281/zenodo.21952872. github.com/Tinonturja/WCF_Biofilm_Adsorption
- **Catkin_Biofilm_Adsorption** — kinetic and isotherm analysis of reactive-dye adsorption on a Saccharum spontaneum cellulose film, with a preregistered simulation of kinetic-law identifiability; modelling archive for a manuscript in preparation. Zenodo DOI 10.5281/zenodo.23195584. github.com/Tinonturja/Catkin_Biofilm_Adsorption
- **tsfabrics_alignment_pilot** — preregistered computer-vision pilot on motion-aligned anomaly detection in knitted-fabric video (TSFabrics); stopped at validation under its own stop rule, test set unscored; code, decision log and technical note public. github.com/Tinonturja/tsfabrics_alignment_pilot

TECHNICAL SKILLS

Languages Python, SQL, C, Java, Bash, LaTeX

Machine learning PyTorch, scikit-learn, NumPy, pandas; deep learning, CNNs, model evaluation and validation

Scientific ML Physics-informed neural networks (inverse problems), Gaussian process regression, nonlinear kinetic/isotherm fitting, molecular dynamics post-processing

Data and tooling Large-scale industrial data extraction and cleaning, relational databases, Git/GitHub, reproducible research (Zenodo), Jupyter

TEACHING

Instructor, Mathematics and Analytical Reasoning — Udvash

Mar 2018 – Jun 2024

Taught mathematics and analytical reasoning to university-admission candidates for six years, part-time alongside study.

STANDARDISED TESTS

GRE General 313/340 — Quantitative 162, Verbal 151, Analytical Writing 3.5 (September 2024)

IELTS Academic 7.5/9.0 — Listening 8.5, Reading 7.5, Writing 7.0, Speaking 7.0 (November 2024)

CERTIFICATIONS

Machine Learning Specialization — DeepLearning.AI and Stanford University (January 2025)

Generative AI Engineering Professional Certificate — IBM (January 2026)

LEADERSHIP

Vice President, Ekattor Cultural Club (2022–23); Joint Secretary, BUTEX Career Club (2021–22); Deputy Chief of Logistics, BUTEX Business Club (2020–22), Bangladesh University of Textiles.